Model selection for the baseline surrogate

This document records how the surrogate used in the rest of the project was chosen, with every number and every calculation needed to check the decision. All runs use `src/train_baseline.py`.

Summary. Starting from the paper's architecture (3 hidden layers × 256 units, no regularisation, validation MSE 0.0571), a deeper network with weight decay reaches **0.0493 ± 0.0008** (5 × 256, weight decay 0.1), about **14 % lower**, far above the run-to-run noise (≈ 0.0009). Dropout and narrower networks do not help. Two configurations are statistically indistinguishable (A: 5 × 256, wd 0.1 and B: 6 × 256, wd 0.2); the simpler one, **A**, is the reference surrogate. B is kept for comparison because it memorises its training data less (train/validation gap ×4.2 vs ×6.3 on average), which may matter for calibration. One target, the minimum normalised magnetic gradient scale length, stays at $R^2 \approx 0.76$ whatever the setting. With 20 networks each (section 7), the validation MSE is 0.0508 (paper), **0.0449 (A)** and 0.0457 (B), and A beats the paper's architecture on 11 of the 12 test metrics.

---

1. Protocol

1.1 Data and split (identical for every run)

- Data: paper baseline subset, **26,746 shapes** (NFP = 3, DESC or VMEC optimisations, 0.05 % tails trimmed per target), built by `src/data.py`.
- Split RAND, data seed 42: **19,258** training shapes, **2,139** validation shapes (10 % of the training part), **5,349** test shapes (20 %). The same shapes are in each set for every run of this document.
- Inputs: the 81 Fourier coefficients that vary; targets: the 12 metrics (QI residual as log10). Both are z-scored with training-set statistics: $z = (y - \mu_{train})/\sigma_{train}$.

1.2 What is fixed

element	value
loss	mean squared error over the 12 z-scored targets
optimiser	AdamW, learning rate $\eta = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$ (PyTorch defaults); with weight decay 0 this is Adam
batch size	256, i.e. $\lceil 19\,258/256 \rceil = 76$ steps per epoch
learning-rate schedule	halved when the validation loss has not improved for 10 epochs
early stopping	stop after 30 epochs without improvement (max 500); keep the best epoch
activation	tanh
ensemble size during the search	$K = 3$ networks per run, network seeds $42 + 100 + 1000\,m + k$ (m = `--member-seed`, $k = 0, 1, 2$)

1.3 What varies

Depth (`--depth`), width (`--width`), dropout (`--dropout`), weight decay (`--weight-decay`).

1.4 Selection criterion

The criterion is the **ensemble validation MSE**: average the predictions of the K networks, $\bar{y} = \frac{1}{K}\sum_k \hat{y}^{(k)}$, then

$$\mathrm{MSE}_{val} = \frac{1}{12\,N_{val}} \sum_{i=1}^{N_{val}} \sum_{j=1}^{12} \left(\bar{y}_{ij} - y_{ij}\right)^2, \qquad N_{val} = 2\,139.$$

- **Not the test scores**: the test set is never used to choose; choosing on it would make the final test score optimistic.
- **Not the train/validation gap** ($\mathrm{MSE}_{val}/\mathrm{MSE}_{train}$): it describes memorisation, but a smaller gap can come with a worse validation error (see dropout below).
- Caveat: the validation set also drives early stopping, so the validation MSE is slightly optimistic. This affects all runs alike and does not bias the comparison.
- Caveat: in this average the gradient scale length weighs about 45 % of the total because its error is by far the largest ($0.51^2 \approx 0.26$ against 0.01–0.06 for the other targets). Section 6 checks that the gains hold on the other 11 targets.

---

2. Search, round by round (one run per setting, member seed 0)

Columns: ensemble MSE on training and validation sets, gap = validation / training, and the minimum test R^2 over the 12 targets (displayed only, never used to choose). Parameter counts include biases.

2.1 Weight decay, first round (scale too small)

run	train	validation	gap	R^2 min
reference 3 × 256	0.0117	0.0571	×4.9	0.736
wd 1e-5	0.0117	0.0571	×4.9	0.736
wd 1e-4	0.0117	0.0571	×4.9	0.736
wd 1e-3	0.0118	0.0571	×4.8	0.736

No effect at all. Explanation: AdamW shrinks every weight by a factor $(1 - \eta\lambda)$ at each step. Over N steps, ignoring the gradients, $\theta_N \approx \theta_0\, e^{-\eta\lambda N}$. With $\eta = 10^{-3}$ and $N \approx 76 \times 100 = 7\,600$ steps (≈ 100 epochs before early stopping):

λ	$\eta\lambda N$	total shrink $1 - e^{-\eta\lambda N}$
1e-5	0.000076	0.01 %
1e-4	0.00076	0.08 %
1e-3	0.0076	0.76 %
1e-2	0.076	7.3 %
5e-2	0.38	32 %
1e-1	0.76	53 %
2e-1	1.52	78 %
5e-1	3.8	98 %

The first round was in the “no effect” zone. Values that do something lie between about 1e-2 and 1e-1, which is also the usual range for AdamW (PyTorch default 1e-2). This is an order of magnitude only: the gradients pull useful weights back, so the real shrink is smaller.

2.2 Dropout

run	train	validation	gap	R^2 min
dropout 0.05	0.0184	0.0592	×3.2	0.755
dropout 0.1	0.0268	0.0643	×2.4	0.753
dropout 0.2	0.0449	0.0732	×1.6	0.743

The gap falls steadily (less memorisation) but the validation error rises: every dropout rate is worse than no dropout (0.0571). Dropout is not used in the reference model (it will be used for MC Dropout later).

2.3 Weight decay, second round (right scale)

run	train	validation	gap
wd 1e-2	0.0122	0.0569	×4.6
wd 5e-2	0.0133	0.0562	×4.2
wd 1e-1	0.0145	0.0550	×3.8
wd 5e-1	0.0349	0.0597	×1.7

A U-shaped curve: the validation error decreases up to 1e-1, then rises at 5e-1, where the training error triples (the network can no longer fit the data, i.e. underfitting).

2.4 Architecture

run	parameters	train	validation	gap
2 × 128	28,556	0.0282	0.0604	×2.1
3 × 128	45,068	0.0219	0.0576	×2.6
5 × 128	78,092	0.0168	0.0549	×3.3
6 × 128	94,604	0.0185	0.0562	×3.0
3 × 256 (paper)	155,660	0.0117	0.0571	×4.9
4 × 256	221,452	0.0077	0.0546	×7.1
5 × 256	287,244	0.0067	0.0531	×7.9
6 × 256	353,036	0.0055	0.0528	×9.5
8 × 256	484,620	0.0056	0.0539	×9.6
3 × 512	573,452	0.0059	0.0563	×9.6

Depth helps up to 5–6 layers, then stops (8 layers); width helps much less (3 × 512); narrow deep networks (× 128) are worse than their 256-unit counterparts. Since reducing capacity never helped and adding depth did, the remaining error is dominated by **bias** (model too simple), not by variance.

2.5 Depth combined with weight decay

run	train	validation	gap
5 × 256, wd 1e-1	0.0077	0.0491	×6.3
5 × 256, wd 2e-1	0.0132	0.0493	×3.7
6 × 256, wd 1e-1	0.0070	0.0493	×7.1
6 × 256, wd 2e-1	0.0110	0.0483	×4.4

The combination gives the largest gain of the study (≈ 0.053 → ≈ 0.049). A plausible reading: a deeper network has more capacity, hence a stronger tendency to memorise, which weight decay counteracts.

3. Measuring the run-to-run noise

3.1 Why `--member-seed` and not `--seed`

A validation score mixes two things: the effect of the setting, and the luck of the random initialisation and batch order. To measure the luck alone, the same setting is retrained with other network seeds on **the same data**.

option	test shapes	validation shapes	network seeds
default (`--seed 42 --member-seed 0`)	draw with 42	draw with 43	142, 143, 144
`--seed 43`	**other shapes**	**other shapes**	143, 144, 145
`--member-seed 1`	same	same	**1142, 1143, 1144**

`--seed` would also change which shapes are scored, mixing the luck of the networks with the difficulty of the validation shapes, and making the scores incomparable with all previous runs. `--member-seed` changes only the networks.

3.2 Formulas

For n runs of one setting with validation scores $x_1, \dots, x_n$:

- mean: $\bar{x} = \frac{1}{n} \sum_i x_i$
- standard deviation (run-to-run noise), with $n-1$ because the mean is estimated from the same data: $s = \sqrt{\frac{1}{n-1} \sum_i (x_i - \bar{x})^2}$
- uncertainty of the mean: $s/\sqrt{n}$
- uncertainty of the difference between two settings A and B: $\sigma_\Delta = \sqrt{s_A^2/n_A + s_B^2/n_B}$
- decision rule: the difference is considered real if $|\bar{x}_A - \bar{x}_B| > 2\,\sigma_\Delta$ (a deviation of more than about two standard deviations happens by chance roughly 5 times in 100 for Gaussian noise); between 1 and 2, the comparison is inconclusive.

The precision of s itself is limited: its relative error is about $1/\sqrt{2(n-1)}$, i.e. ≈ 50 % for $n = 3$ and ≈ 35 % for $n = 5$. These measurements give the order of magnitude of the noise, which is enough to separate clear cases from unclear ones.

3.3 Three runs per setting (member seeds 0, 1, 2)

A = 5 × 256, wd 1e-1: 0.0491, 0.0507, 0.0488.

- $\bar{x}_A = (0.0491 + 0.0507 + 0.0488)/3 = 0.04953$
- deviations: −0.00043, +0.00117, −0.00073; sum of squares 2.087×10^{-6}
- $s_A = \sqrt{2.087 \times 10^{-6}/2} = 0.00102$

B = 6 × 256, wd 2e-1: 0.0483, 0.0491, 0.0476.

- $\bar{x}_B = 0.04833$
- deviations: −0.00003, +0.00077, −0.00073; sum of squares 1.127×10^{-6}
- $s_B = \sqrt{1.127 \times 10^{-6}/2} = 0.00075$

Comparison

- difference: $0.04953 - 0.04833 = 0.00120$
- $\sigma_\Delta = \sqrt{0.00102^2/3 + 0.00075^2/3} = 0.00073$
- ratio $0.00120/0.00073 = 1.64$: **inconclusive** (Welch t-test p ≈ 0.18).

3.4 Five runs per setting (member seeds 0 to 4)

A: 0.0491, 0.0507, 0.0488, 0.0489, 0.0492.

- $\bar{x}_A = 0.04934$; deviations −0.00024, +0.00136, −0.00054, −0.00044, −0.00014; sum of squares 2.412×10^{-6}; $s_A = \sqrt{2.412 \times 10^{-6}/4} = 0.00078$

B: 0.0483, 0.0491, 0.0476, 0.0478, 0.0499.

- $\bar{x}_B = 0.04854$; deviations −0.00024, +0.00056, −0.00094, −0.00074, +0.00136; sum of squares 3.652×10^{-6}; $s_B = \sqrt{3.652 \times 10^{-6}/4} = 0.00096$

Comparison

- difference: $0.04934 - 0.04854 = 0.00080$
- $\sigma_\Delta = \sqrt{0.00078^2/5 + 0.00096^2/5} = 0.00055$
- ratio $0.00080/0.00055 = 1.45$: **inconclusive** (Welch t-test p ≈ 0.19).

With more runs, the difference shrank from 0.00120 to 0.00080 while its uncertainty shrank less: the behaviour expected when an apparent difference is mostly chance. B is lower than A for 4 seeds out of 5, but the gap is about 1.6 % of the validation error and reverses for seed 4.

3.5 Noise level used in the rest of the document

Pooling the two settings (5 runs each): $s = \sqrt{(4 \times 0.00078^2 + 4 \times 0.00096^2)/8} = 0.00087$. We round up to $\sigma \approx 0.0010$ to stay conservative (the larger of the two estimates).

---

4. Re-reading the single-run comparisons with this noise

For two settings run **once** each, each score carries its own noise, so $\sigma_\Delta = \sqrt{\sigma^2 + \sigma^2} = \sqrt{2}\,\sigma \approx 0.0014$, and the threshold is $2\,\sigma_\Delta \approx 0.0028$.

comparison	from (validation)	to (validation)	difference	verdict
add depth	3 × 256: 0.0571	5 × 256: 0.0531	0.0040	real
add weight decay to the deep network	5 × 256: 0.0531	5 × 256, wd 1e-1: 0.0491	0.0040	real
paper → best model	3 × 256: 0.0571	B, mean of 5 runs: 0.0485	0.0086	clearly real
weight decay alone on the paper network	3 × 256: 0.0571	3 × 256, wd 1e-1: 0.0550	0.0021	within noise
5 vs 6 layers (no weight decay)	5 × 256: 0.0531	6 × 256: 0.0528	0.0003	within noise

For "paper → best model" one side is a mean of 5 runs, so $\sigma_\Delta = \sqrt{0.0010^2 + 0.0010^2/5} \approx 0.0011$ and the threshold is ≈ 0.0022; the difference is about 8 times σ_Δ.

---

5. Decision

- **Reference surrogate: A = 5 × 256, weight decay 0.1, no dropout.** A and B are indistinguishable (section 3.4), so the simpler model is chosen: 287,244 parameters instead of 353,036, faster to train, and the better-measured of the two ($s_A = 0.00078$).

- **B = 6 × 256, weight decay 0.2 is kept for comparison.** Same accuracy, but it memorises less (gap ×4.2 vs ×6.3 on average over the 5 runs: A 6.3, 5.6, 6.1, 6.8, 6.8; B 4.4, 4.2, 4.2, 4.2, 4.0). The uncertainty study will check whether stronger memorisation makes the ensemble over-confident.
- **The paper's architecture (3 × 256, no regularisation) is kept as a faithful reproduction.**

Why A uses weight decay 0.1 and B uses 0.2. These values come from single runs (section 2.5): for 5 × 256, 0.1 scored 0.0491 and 0.2 scored 0.0493; for 6 × 256, 0.1 scored 0.0493 and 0.2 scored 0.0483. The numerically better value was taken for each depth, but both differences (0.0002 and 0.0010) are below the single-run threshold of 0.0028. The choice of weight decay within [0.1, 0.2] is therefore **not demonstrated**, and A and B differ by two settings at once (depth and weight decay). This does not affect the conclusion that A and B are equivalent, but it does not allow attributing any difference between them to depth alone.

---

6. Check on all 12 targets

Test RMSE (z-scored) and test R^2 per target for the three final models, each an ensemble of $K = 20$ networks (section 7). This check uses the test set to verify that the validation criterion is sensible, not to choose between settings: it was looked at once, after the decision of section 5.

target	paper RMSE	A RMSE	B RMSE	paper R^2	A R^2	B R^2
aspect ratio	0.102	0.087	0.083	0.990	0.992	0.993
aspect ratio / edge iota	0.240	0.223	0.221	0.936	0.945	0.946
max elongation	0.256	0.223	0.220	0.936	0.951	0.952
axis iota / NFP	0.202	0.187	0.191	0.954	0.961	0.959
edge iota / NFP	0.140	0.129	0.130	0.978	0.981	0.981
axis mirror ratio	0.156	0.135	0.132	0.976	0.982	0.983
edge mirror ratio	0.135	0.119	0.118	0.982	0.986	0.986
average triangularity	0.169	0.153	0.156	0.971	0.976	0.975
vacuum well	0.256	0.236	0.240	0.936	0.945	0.943
min. magnetic gradient scale length	**0.492**	**0.490**	**0.493**	**0.757**	**0.759**	**0.756**
flux compression in bad curvature	0.214	0.200	0.201	0.955	0.960	0.960
log10 qi	0.201	0.185	0.190	0.960	0.966	0.964

- **A vs paper:** 11 targets improve by 6.5 to 15 % in RMSE (aspect ratio: $(0.102-0.087)/0.102 = 15\,\%$; edge iota / NFP: $(0.140-0.129)/0.140 = 8\,\%$), and the gradient scale length is unchanged (0.492 → 0.490). The validation criterion and the test set agree.

- **A vs B:** differences of at most 0.005 in RMSE, in both directions (B better on 5 targets, A on 7): consistent with the two models being equivalent.
- **The gradient scale length** stays at $R^2 \approx 0.76$ for every model and every setting of the search (0.73 to 0.76). Its error is probably dominated by noise or by information absent from the boundary coefficients (it is a **minimum** over the surface, which can jump when the location of the minimum moves), not by the model.
- **Gap with the paper's reported accuracy** ($R^2 > 0.97$ for all metrics): without the gradient scale length, the lowest R^2 is 0.936 (paper reproduction) and 0.945 (A); 3 targets of A remain below 0.96 (aspect ratio / edge iota, max elongation, vacuum well).

---

7. Final models

Commands (20 networks each, member seed 0, same split and validation set as the search):

```
uv run python src/train_baseline.py --n-members 20 --depth 5 --width 256 --weight-decay 1e-1 --tag _A_final
uv run python src/train_baseline.py --n-members 20 --depth 6 --width 256 --weight-decay 2e-1 --tag _B_final
uv run python src/train_baseline.py --n-members 20 --tag _paper
```

Reproducibility. Networks 0 to 9 of `_paper` stop at exactly the same epochs (114, 99, 104, 91, 99, 91, 113, 96, 109, 119) with the same validation MSE as the first 10-network run of the paper's architecture: same data seed, same network seeds and same code give the same numbers on CPU.

7.1 Results

model	params	epochs	train	val. (K = 20)	val. 1 network	gap	test R^2 min	test RMSE max
paper	155,660	81–119	0.0109	0.0508	0.0748	×4.7	0.757	0.492
A	287,244	98–185	0.0071	**0.0449**	0.0619	×6.3	0.759	0.490
B	353,036	169–228	0.0109	0.0457	0.0577	×4.2	0.756	0.493

Paper = 3 × 256, A = 5 × 256 with weight decay 0.1, B = 6 × 256 with weight decay 0.2. All MSE are z-scored and averaged over the 12 targets; "val. 1 network" is the mean over the 20 networks of each network's own validation MSE; epochs per network; gap = validation / train ensemble MSE, e.g. A: $0.0449/0.0071 = 6.3$. Total training time on a laptop CPU: about 19 min (paper), 57 min (A) and 62 min (B). Regularised networks train longer because weight decay slows down memorisation, so the validation minimum comes later.

- **A vs paper:** $0.0508 - 0.0449 = 0.0059$, i.e. 12 % lower. At $K = 3$ the run-to-run noise was $\sigma \approx 0.0010$ and it can only be smaller with 20 networks: the gain is real.
- **A vs B:** $0.0457 - 0.0449 = 0.0008$, now in favour of A (it was in favour of B at $K = 3$, section 3.4). The sign flips with the networks drawn: A and B remain equivalent, as concluded in section 5.

7.2 How the error depends on the number of networks

For an ensemble of K networks trained independently, the expected MSE splits into a part the average cannot remove and a part that shrinks with K:

$$\mathrm{MSE}_K = a + \frac{b}{K}$$

- a: bias of the architecture and noise in the data (what all networks get wrong together);

- b: variance between networks (where they disagree); averaging K networks divides it by K.

Fitting a and b from two points. $K = 1$ is the mean validation MSE of a single network, $K = 20$ the ensemble:

$$a + b = \text{MSE}_1, \qquad a + \frac{b}{20} = \text{MSE}_{20} \quad \Rightarrow \quad b = \frac{\text{MSE}_1 - \text{MSE}_{20}}{1 - 1/20}, \qquad a = \text{MSE}_1 - b.$$

- **A:** $b = (0.0619 - 0.0449)/0.95 = 0.0179$; $a = 0.0619 - 0.0179 = 0.0440$.
- **B:** $b = (0.0577 - 0.0457)/0.95 = 0.0127$; $a = 0.0577 - 0.0127 = 0.0450$.
- **paper:** $b = (0.0748 - 0.0508)/0.95 = 0.0253$; $a = 0.0748 - 0.0253 = 0.0495$.

Check against the $K = 3$ runs of the search (prediction $a + b/3$ vs measurement):

model	prediction $a + b/3$	measured at $K = 3$
A	$0.0440 + 0.0179/3 = 0.0500$	0.0493 ± 0.0008 (5 seeds)
B	$0.0450 + 0.0127/3 = 0.0492$	0.0485 ± 0.0010 (5 seeds)
paper	$0.0495 + 0.0253/3 = 0.0579$	0.0571 (1 run)

All three predictions fall within about one noise level of the measurements: the model $a + b/K$ describes the data well.

What it shows.

- Beyond 20 networks the error can hardly decrease: for A, $b/20 = 0.0009$ is all that is left to gain, and MSE_K never goes below $a = 0.0440$. The remaining error is bias (or noise in the data), not variance, as concluded in section 2.4.
- The regularised models have a lower floor a (0.044–0.045 vs 0.0495): this is where their gain over the paper comes from.
- B's networks are individually better than A's (0.0577 vs 0.0619) but **disagree less** (b = 0.0127 vs 0.0179), so averaging helps B less and both ensembles end at the same level. Since the ensemble's uncertainty is precisely the disagreement between networks, B should produce **narrower** uncertainty intervals than A. Whether they are too narrow (over-confident) is the first question of the uncertainty study.

The first 10 networks of each run form the $K = 10$ ensembles compared with the paper; all 20 are used to study how calibration depends on K.

---

8. Limitations

- Noise was measured on 2 settings with 5 runs each; its own uncertainty is about ±35 %.
- About 25 settings were compared on the same validation set: the best ones benefit slightly from the luck of that set. The test set, untouched, gives the honest final score.
- The search covered depth, width, dropout and weight decay only; learning rate, batch size and activation were not tuned, and the paper's own tuned values are not published.
- All results are on the RAND split (in-distribution).